

Avinash Thadani, Ph.D.

EDUCATION

University of Toronto Doctor of Philosophy (Chemistry), 2001	Toronto, Ontario
University of Toronto Master of Science (Chemistry), 1998	Toronto, Ontario
University of Toronto Bachelor of Science (Hons.), Biochemistry, 1996	Toronto, Ontario

ACADEMIC APPOINTMENTS

Georgian College School of Health, Wellness and Sciences Nursing Degree Program (BScN), Pre-Health Sciences	December 2011 - present Faculty – Full time
University of Windsor, Assistant Professor Department of Chemistry & Biochemistry Bachelor of Science Program	July 2004 - November 2011 Faculty – Full time
University of Chicago, NSERC Post-Doctoral Fellow Department of Chemistry	July 2001 – June 2004

NON-ACADEMIC EMPLOYMENT

Thadani Consulting Services	July 2004 – June 2016
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Consultant for the development of chemical and biochemical process for the efficient synthesis of precursor molecules for pharmaceutical agents. The main client was Kanata Chemical Technologies Inc. As a result of the consulting work, several new technologies were developed, and five (5) American patents were filed:

US Patent 7230125 “Methods of Performing Cycloadditions, Reaction Mixtures, and Methods of Performing Asymmetric Catalytic Reactions”.
Inventors: Rawal, Viresh H; Huang, Yong; Unni, Aditya K; Thadani, Avinash N.

- US Patent 8143451 “Methods of Preparing Tertiary Carbinamine Compounds”.
Inventors: Thadani, Avinash N.; Dhudshia, Bhartesh.
- US Patent 8350047 “Methods of Preparing Secondary Carbinamine Compounds with Boronic Acids”. *Inventors:* Thadani, Avinash N.; Dhudshia, Bhartesh
- US Patent 8383859 “Methods of Preparing Primary, Secondary and Tertiary Carbinamine Compounds in the Presence of Ammonia”.
Inventors: Thadani, Avinash N.; Dhudshia, Bhartesh
- US Patent 8759541 “Chiral Acyclic Diaminocarbene Ligands, Precursors Therefore and Their Use in Organic Synthesis Reactions”
Inventors: Beshai, Mira; Thadani, Avinash N.; Ntaganda, Rukundo

ADMINISTRATIVE APPOINTMENTS

Chair, Georgian College Research Ethics Board January 2023 – present

Overseeing and guiding the overall operations of the Georgian College Research Ethics Board (GCREB) including the submission, review, and revision of the ethics of course based and regular research applications with human participants in accordance with the guidelines of the TCPS2.

Member, Georgian College Research Ethics Board Sep. 2021 – December 2022

Reviewed applications for research ethics approval; designed learning resources for potential applicants

TEACHING EXPERIENCE

Honours Bachelor of Science in Nursing program, Georgian College

NURS 1021 Pathophysiology 2, 2022 - present

NURS 1020 Anatomy & Physiology 2, 2022 - present

NURS 1017 Pathophysiology 1, 2022 - present

NURS 1016 Anatomy & Physiology 1, 2022 - present

Pre-Health program

CHEM 1008 Chemistry for Health and Science 1, 2013 - 2016

CHEM 1009 Chemistry for Health and Science 2, 2014 - 2016

Honours Bachelor of Science program, University of Windsor

CHEM 2300 Introductory Organic Chemistry, 2004 – 2010

CHEM 2310 Introductory Organic Chemistry II, 2004 – 2010

CHEM 3310 Intermediate Organic Chemistry, 2004 – 2010

CHEM 4308 Special Topics in Organic Chemistry, 2004 – 2010

CHEM 4900 Research, 2004 – 2010

Masters of Science program, University of Windsor
 CHEM 8328 Advanced Topics in Organic Chemistry, 2004 – 2010
 CHEM 8970 Master's Thesis, 2004 – 2010

Doctor of Philosophy program, University of Windsor
 CHEM 9980 Doctoral Dissertation, 2004 – 2010

SCHOLARSHIP

SCHOLARSHIP OF DISCOVERY

Research Studies (2024-2017 only)

Using Virtual Reality to Teach Anatomy: Can It Help Improve Motivation and Learning Gains in College Students; Co-investigator. Funded by Future Skills Canada (FSC)
 Shock-Proofing the Future of Work: Skills Innovation Challenge Grant (2021 – 2023)

Publications (2024 – 2017 only)

Thadani, A., Deschamps, I., Doran, J., Forlani, C., Theriault, R., & Madorin, S. (2025). Student Motivation Using Virtual Reality in Human Anatomy and Physiology Courses. *Canadian Journal of Learning and Technology*, 51(2), 1–21. <https://doi.org/10.21432/cjlt28609>

SCHOLARSHIP OF TEACHING

Thadani, A. (2024) Development of an online for the delivery of customized next generation NCLEX questions for RN students.

- A web-based platform was developed through JavaScript programming to deliver tailored, specific NGN questions in the A&P and Pathophysiology courses in the HBSN program

DEVELOPMENT OF CURRICULUM, COURSES, AND COURSE MATERIALS

Curriculum

Development of the HBSN curriculum at Georgian College using an integrated and unified framework. Program learning outcomes were carefully created, and specific course-based learning outcomes were subsequently developed to support them. The program was approved by the CNO and successfully launched in Fall 2022.

Academic Courses Developed (2024 – 2017 only)

NURS 1016 Anatomy & Physiology 1

Developed the course and all associated learning resources and assessment based on an open educational resource (OER) textbook from OpenStax.

NURS 1020 Anatomy & Physiology 2

Developed the course and all associated learning resources and assessment based on an OER textbook from OpenStax.

NURS 1017 Pathophysiology 1 (in-progress)

Redeveloping the course and all associated learning resources and assessment based on publicly available, reliable, and validated data sources.

NURS 1021 Pathophysiology 2 (in-progress)

Redeveloping the course and all associated learning resources and assessment based on the publicly available, reliable, and validated data sources.

NURS 2020 Advanced Pathophysiology and Microbiology (in-progress)

Developing the course and all associated learning resources and assessment based on the publicly available, reliable, and validated data sources as well as the use of an OER textbook for Microbiology

PROFESSIONAL MEMBERSHIPS

Canadian Society of Chemistry

American Society of Chemistry

PROFESSIONAL DEVELOPMENT AND CONTINUING EDUCATION (2024 – 2017 only)

Course on Research Ethics based on the Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS 2: CORE 2022)

Panel of Research Ethics 2022

VR educator Micro-credential Course

Georgian College 2021

The Curator and Technologist Micro-credential Courses

Ontario Extend 2020

The Keystones of Healthcare Simulation Certificate Program

SIM-one, Healthcare Simulation Network 2017